

# Reading the economy from space

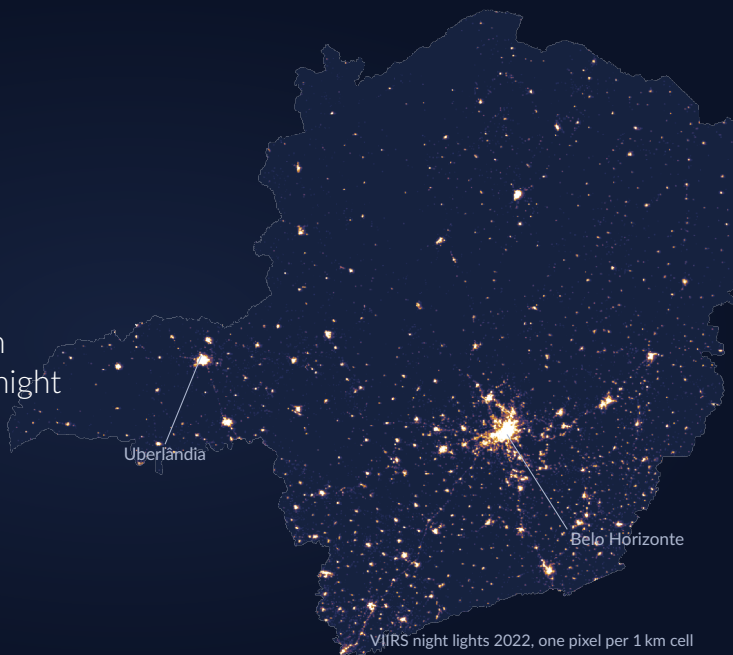
Estimating municipal GDP from satellite embeddings, text and night lights

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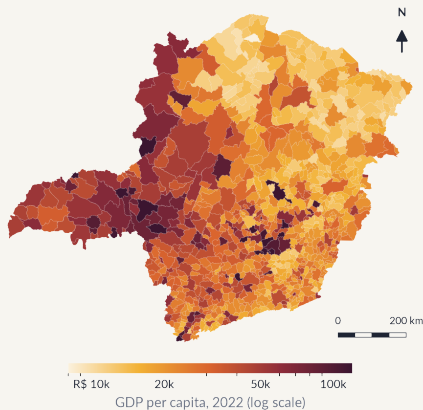
**Hemant** (22CS30029)

Minas Gerais, Brazil · 853 municipalities

October 2026



# Local GDP is patchy; satellites see every place



49×

gap in GDP per capita inside one state:

**Catas Altas (R\$ 391k) vs São João das Missões (R\$ 8k)**

- Sub-national GDP is published **late**, **estimated** from partial data, or **missing** where statistics are weak.
- Satellites measure **every place, every year, with the same sensor**.

## GOAL

Estimate GDP per capita for all **853 municipalities** of Minas Gerais from **free satellite data**, and test it **honestly**.

# Two recent ideas, never tested together

## UrbanCLIP Yan et al., WWW 2024

A language model writes a **caption for every satellite tile**; image and text are aligned.

- GDP gain only +0.5%  $R^2$ ; random split lets neighbours leak into the test

## Satellite embeddings Yue, Zhao & Hu, 2026

Google's **64-number embeddings, steered by night lights** towards economic meaning.

- countries only; no text, no POI or roads

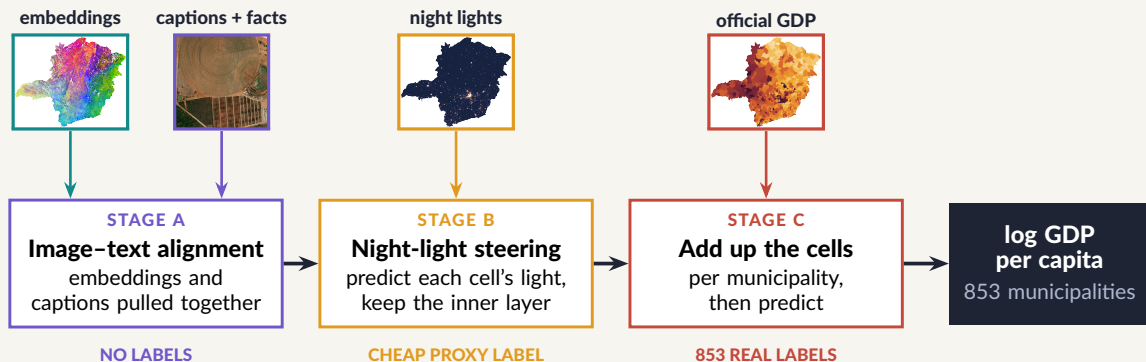
## THIS PROJECT

**Both, in one pipeline**

- embeddings ● text
- night lights ● POI, roads

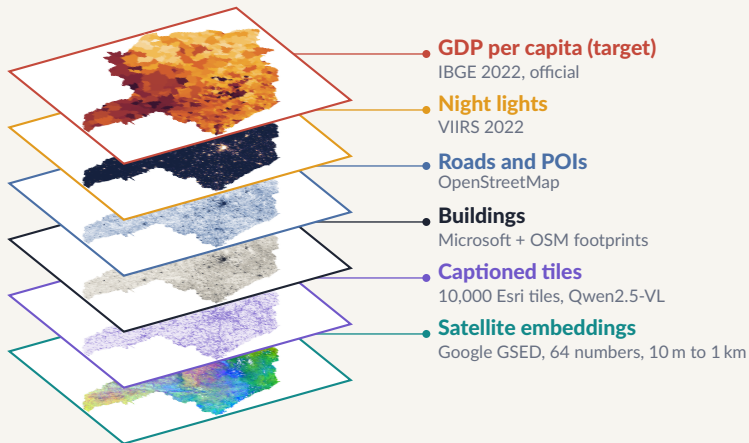
853 municipalities · spatial test  
ablation of every ingredient

# Three learning stages, from no labels to official GDP



Captions are needed only in training: at prediction time each cell uses its embedding alone. Switching stages off gives the ablations (variants V3-V9).

# Six free layers on one 1 km grid



**853**

municipalities

**575k**

1 km cells

**10,000**

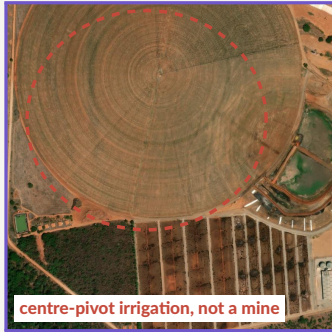
captioned tiles

**10M**

building footprints

All layers free and aligned on the same grid (EPSG:31983). The target is official statistics, not night lights.

# An AI model describes each tile; map data vetoes false claims



## QWEN2.5-VL-7B CAPTION, AFTER THE CHECKS

- ✗ The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland: **no industry in the data**
- ✓ Roads crisscross the area, connecting the plots and the perimeter.
- ✓ Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- ✓ A small body of water, possibly a pond, is visible near the farmland.
- ✓ No buildings or industrial structures are evident, suggesting minimal built-up area.
- ✗ The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations: **filler**

Esri World Imagery, about 1 m per pixel

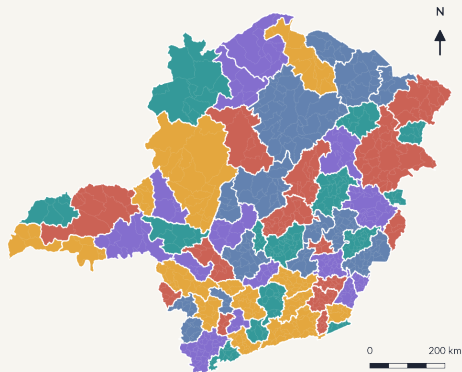
## SENTENCES REMOVED, BY RULE

|                    |       |
|--------------------|-------|
| Filler, no content | 4,183 |
| Industry / mine    | 1,130 |
| Building           | 1,108 |
| Water              | 1,046 |
| Road               | 1,039 |
| Farmland           | 732   |

**84%** kept of 60,023 sentences

Checked against ESA land cover, OSM roads and building footprints; image-text matching removes 246 more captions. A blind hand audit is in progress.

# Every municipality is tested in a region never seen



70 official regions (white outlines) grouped into 5 folds

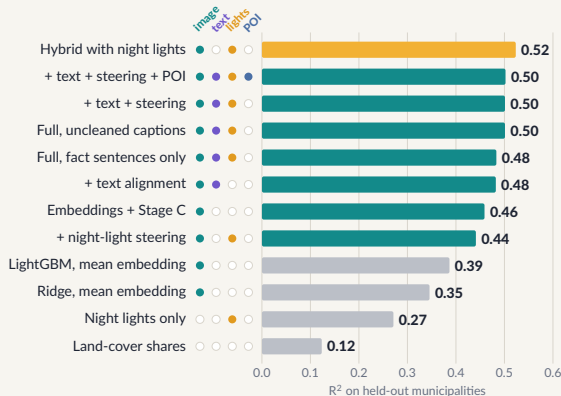
|         | F1       | F2       | F3       | F4       | F5       |
|---------|----------|----------|----------|----------|----------|
| Round 1 | train    | validate | train    | train    | train    |
| Round 2 | train    | test     | validate | train    | train    |
| Round 3 | train    | train    | test     | validate | train    |
| Round 4 | train    | train    | train    | test     | validate |
| Round 5 | validate | train    | train    | train    | test     |

Legend:  train     validate     test

- Whole regions held out: **no look-alike neighbour** in training. All stages retrained each round.
- UrbanCLIP split tiles **at random**, so neighbours sat on both sides and scores were inflated.

Models are compared on the same folds with a block bootstrap over regions (95% intervals).

# Text gives the biggest gain; the hybrid explains half of GDP



Dots: ingredients each model uses. Hybrid = best model blended with night lights.

**0.52**  $R^2$ , hybrid, unseen regions

**RQ2 TEXT** largest gain, borderline

$\Delta RMSE$  -0.025, 95% CI [-0.050, +0.004]

**RQ1 STEERING** no clear effect

$\Delta RMSE$  +0.007, 95% CI [-0.028, +0.041]

**RQ3 POI, ROADS** nothing added

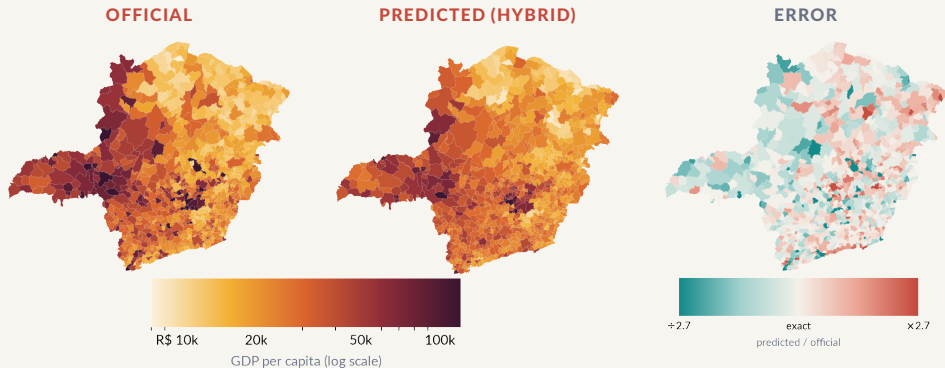
$\Delta RMSE$  -0.001, 95% CI [-0.017, +0.016]

**RQ4 HYBRID** best, not significant

$\Delta RMSE$  -0.009, 95% CI [-0.035, +0.015]

RMSE differences on log GDP per capita with 95% block-bootstrap intervals; negative = first model better.

# Predictions follow the map, but are pulled towards the middle



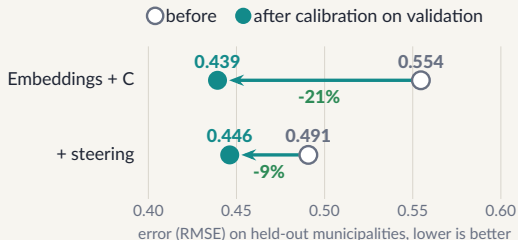
**75%**

of municipalities predicted  
within a factor 1.5 of official

**Under-predicted:** rich agribusiness and mining towns (west, centre).

**Over-predicted:** the poorer north-east. Explaining these errors is RQ5, next.

# Calibration mattered more than night-light steering



- 1 Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by **-21%**.
- 2 Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
- 3 Captions need checking.** The model invents mines, roads and farmland on plain tiles; the data checks remove most of it.

# About 85% done: every experiment has finished



## RESEARCH QUESTIONS

|                          |          |
|--------------------------|----------|
| RQ1 night-light steering | ANSWERED |
| RQ2 text (captions)      | ANSWERED |
| RQ3 POIs and roads       | ANSWERED |
| RQ4 hybrid               | ANSWERED |
| RQ5 where it fails       | NEXT     |

## NEXT

caption audit → error analysis and 1 km maps (RQ5) → report

SO FAR

# Satellite embeddings, captions and night lights, tested honestly, explain half of local GDP.

**0.52**

$R^2$ , hybrid, unseen regions

**0.27**

$R^2$ , night lights alone

**—21% 84%**

error after calibration

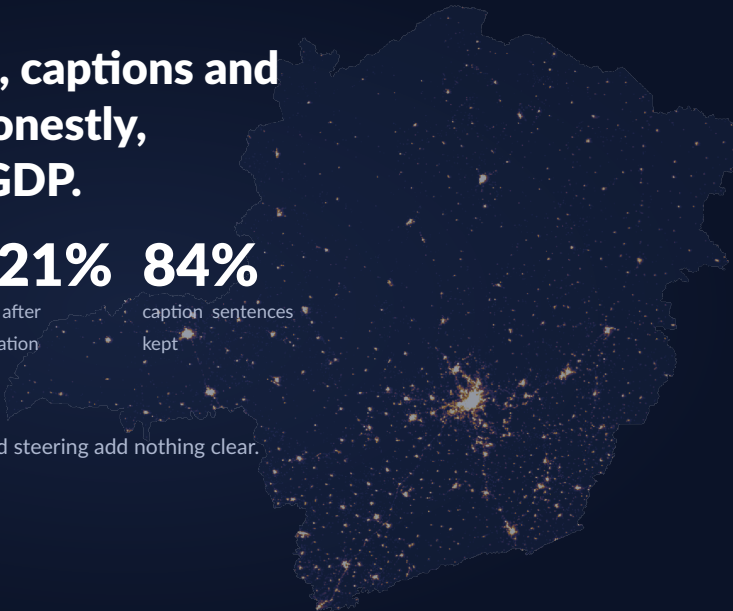
caption sentences kept

**So far:** captions give the largest gain, POIs and steering add nothing clear.

**Next:** where does the model fail (RQ5)?

## Questions?

[github.com/chadsaras/GIS\\_Project](https://github.com/chadsaras/GIS_Project)

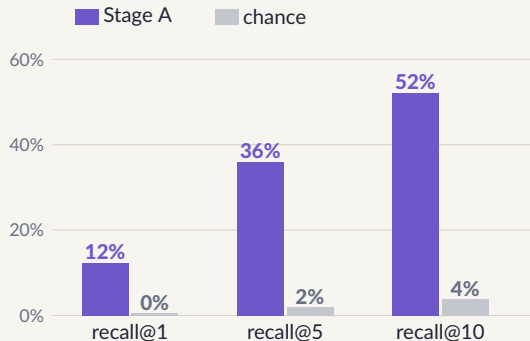


# Data sources

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| Layer                  | Source                                           | Use                                        |
|------------------------|--------------------------------------------------|--------------------------------------------|
| ● Satellite embeddings | Google Satellite Embedding V1 (AlphaEarth), 2022 | image input, averaged 10 m to 1 km         |
| ● Night lights         | VIIRS DNB annual V2.2, 2022                      | Stage B proxy label, night-light benchmark |
| ● Land cover           | ESA WorldCover v200, 2021                        | masking, fact sentences, caption checks    |
| ● Roads, POIs          | OpenStreetMap (Geofabrik, 2023-01-01)            | features, fact sentences, caption checks   |
| ● Buildings            | Microsoft ML footprints + OSM                    | caption checks                             |
| ● Caption imagery      | Esri World Imagery (ArcGIS Location Platform)    | input to the vision-language model         |
| ● GDP, population      | IBGE municipal GDP 2022, Census 2022             | target: log GDP per capita                 |

# Stage A aligns embeddings and captions far above chance



For a validation tile, how often is its own caption among the top  $k$  of 256 candidates?

Part of the alignment comes from the fact sentences; variants V8 (raw captions) and V9 (facts only) separate the two.

# What the captions look like



## FOREST

A predominantly forested area with patches of cleared land.



## FARMLAND

A large, circular area of brown farmland, likely recently plowed, dominating the center.



## VILLAGE EDGE

A sparsely built area with scattered houses and small buildings, likely residential.



## TOWN CENTRE

A densely built-up area with numerous houses featuring red-tiled roofs, indicating a residential neighborhood.

First sentence of each caption that passed the checks.